

Mukul Hase

Senior Software Engineer · AI Systems & Desktop Platforms
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Senior software engineer with eight years building AI systems, desktop platforms, and edge computer vision products. Comfortable going a layer below the problem, into inference infrastructure, OS internals, or the build system, and seeing the work through to what users actually touch.

EXPERIENCE

- **Figma** San Francisco, CA
Senior Software Engineer Sep 2021 – Current
 - **Figma Agent, agent platform & AI design editing (2025–Current):** Worked on core platform behind Figma’s AI agent: Agent orchestration across Claude, Codex, and OpenCode runtimes, sandboxed agent execution, MCP tool interfaces, streaming protocols with disconnect and completion semantics, and agent prompt, configuration, and release management. Integrated a Cerebras-hosted in-house model that cut generation from minutes to 5 seconds (30x faster, now serving 80% of traffic) and built the model routing, picker, and fallback layer behind it. Improved reliability across a nondeterministic system: reduced the top failure mode from 4% to 0.5% and auto-recovered 37% of common errors through retries and self-healing agents. Took prompt-to-edit, one surface on that platform, from research prototype to production, including the on-canvas editing surface and undo/commit flow. Built the team’s debug tooling and eval infrastructure. Worked across the stack: React surfaces, TypeScript and Go services, C++ in the editor core, and Python eval tooling.
 - **First Draft, prompt-to-design generation (2024):** Moved to the AI generation team and owned First Draft through general availability. Built the design-system intelligence behind it: matching generated output to a team’s existing component library, detecting IP conflicts, and inferring component props, plus prompt suggestions and progressive loading for slow image generation. Instrumented the analytics and dashboards used to measure generation quality.
 - **Figma Desktop, Electron (2021–2024):** Built the desktop notification feed end to end (macOS/Windows, spanning native, Node, and web layers); tray surface, push delivery, and comment previews lifted notification click-through 25%. Integrated cross-platform native spellcheck, bringing 30+ languages to Figma and FigJam. Rebuilt desktop CI/CD from scratch: signed dogfood and per-PR build channels that caught regressions before they reached beta. Shipped the desktop side of Figma’s UI3 redesign, launched at Config, plus trackpad haptics, light mode, and the tab bar.
- **Microsoft** Redmond, WA
Software Engineer II June 2018 – Sep 2021
 - **Whiteboard sharing feature, Microsoft Teams:** Built UI/UX in the Teams desktop app (React, GraphQL, Electron, Apollo).
 - **Edge computer vision for document cleanup, cropping & classification:** Built a parametric image rendering pipeline generating realistic synthetic document images for training and scoring, Python tooling for data collection/filtering/processing, a data-collection and training pipeline on AML, and voting/annotation/analysis tools. Ported models to ONNX/CoreML/TFLite and ran them natively on Android and iOS.
 - **Native mobile development, Office Lens (Android & iOS):** Camera stack, Kotlin re-architecture, GPU shaders (Metal & OpenGL), custom UI animations, and a 10x improvement in document deletion/update time.
- **FreeAgent Software** Saratoga, CA
SWE Intern (Bay Area startup) May – July 2017
 - **Full-stack web development:** React, GraphQL, Express. Built a merge feature, led an incremental-ID to UUIDv4 key migration, and built a filtering sidebar with multi-property constraints across query APIs and UI.

PROJECTS

- **WhatsAPI:** Python library for WhatsApp Web, 1500+ GitHub stars, deployed on PyPI. [github]
- **Others:** Pebble Messenger (Shake2talk over Facebook chat, 260 users); 3-Sweep (Photoshop-style 2D-to-3D image manipulation) [github]; WebGL 3D Game; IIIT’s event portal (personalized calendars by student “groups,” integrated with college LDAP).

AWARDS

- 2018 · Dean’s List, IIIT, 4th Year
- 2017 · 1st Place, NBO Hackathon
- 2016 · 2nd Place, Microsoft Code.Fun.Do
- 2016 · Dean’s List, IIIT, 2nd Year
- 2015 · Dean’s List, IIIT, 1st Year

EDUCATION

- **IIIT Hyderabad** · B.Tech Computer Science · CGPA 8.34/10 · 2014–2018

SKILLS

- **Languages:** TypeScript/JavaScript, Python, Go, C++/Objective-C++, Kotlin, Swift, Rust
- **AI & Agent Systems:** LLM agent orchestration, MCP tooling (Claude, Codex, OpenCode), sandboxed multi-agent execution, SSE streaming & chunked responses, model routing & fallback (SageMaker, Cerebras, Fireworks, Anthropic), eval harnesses & observability
- **Desktop Platform:** Electron internals, native macOS/Windows (AppKit, Obj-C++, C++), code signing & notarization (Apple, DigiCert HSM), release engineering
- **Web, Backend & Infra:** React, GraphQL, Node/Express, Protobuf, GitHub Actions, Bazel, Terraform, AWS, Datadog/Sentry